Factory-to-Customer Shipping Route Efficiency Analysis

Nassau Candy Distributor

Professional Project Report

Logistics Analytics & Route Performance Study

Prepared by	Nikhil Chikte
Project Type	Data Analytics / Supply Chain Analysis
Primary Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter
Project Status	Analysis Completed

This report documents the analytical approach, key results, business interpretation, and limitations of the completed Nassau Candy shipping-route analysis.

Abstract

This project evaluates factory-to-customer shipping patterns for Nassau Candy Distributor using order, shipment, customer, product, sales, and route information. The analysis focuses on shipping lead time, regional shipment distribution, shipping-mode performance, and factory-to-customer route efficiency. A structured workflow was applied covering data validation, feature engineering, route aggregation, KPI analysis, geographic analysis, and visualization. The working dataset contains 10,194 shipment records. Regional analysis shows the Pacific region has the largest shipment volume, followed by Atlantic, Interior, and Gulf. Standard Class represents the largest shipment-mode category. A significant data-quality limitation was identified: calculated shipping lead times are unusually high, so the values should be validated before being interpreted as real-world operational benchmarks.

1. Introduction

Efficient logistics operations depend on understanding how orders move from origin facilities to customers. Shipping-route analysis can reveal differences in shipment volume, service modes, geographic performance, and potential bottlenecks. For a distributor, these insights can support better operational monitoring and help prioritize routes for further investigation.

The objective of this study is not to redesign the complete logistics network, but to establish a reproducible analytical view of factory-to-customer shipping performance using the available dataset. The resulting analysis provides a foundation for future dashboarding, route optimization, and predictive logistics work.

2. Problem Statement

The project addresses the need to understand shipping-route efficiency across different geographic regions and shipment modes. Without structured analysis, high-volume routes, regional differences, and unusual shipping-time patterns can be difficult to identify.

3. Objectives

- Analyze shipping lead times between order and shipment dates.
- Evaluate factory-to-customer route performance.
- Compare shipment distribution across regions.
- Analyze the use and performance of different shipping modes.
- Identify high-volume routes and potential geographic bottlenecks.
- Calculate logistics and business KPIs including shipments, sales, units, and lead time.
- Document data-quality limitations that affect operational interpretation.

4. Dataset Description

The dataset contains order-level records covering customer, product, sales, shipment, and geographic information. Key fields used in the analysis include Order ID, Order Date, Ship Date, Ship Mode, Customer ID, City, State/Province, Region, Product ID, Product Name, Sales, Units, Gross Profit, Cost, factory-region information, factory-state information, and route-related features.

Major analytical fields

Category	Representative fields
Order / Shipment	Order ID, Order Date, Ship Date, Ship Mode

Category	Representative fields
Customer / Geography	Customer ID, City, State/Province, Region
Product / Financial	Product ID, Product Name, Sales, Units, Gross Profit, Cost
Route Features	Factory Region, Factory State, Shipping Lead Time, Factory-to-Customer Route

5. Methodology

5.1 Data Cleaning & Validation

The dataset structure, data types, missing values, duplicate records, date fields, and shipping-time validity were reviewed before analysis.

5.2 Feature Engineering

Shipping lead time was derived as Ship Date minus Order Date. Factory and customer geographic attributes were combined to support route-level analysis.

5.3 Route Aggregation

Records were grouped by route and geographic dimensions to calculate shipment volume, average lead time, sales, and units.

5.4 Efficiency Benchmarking

Regions, routes, and shipping modes were compared using common KPIs to identify relative differences and areas requiring further investigation.

5.5 Geographic Analysis

Regional and state-level summaries were used to identify high-volume areas and unusual shipping-time patterns.

5.6 Visualization

Charts were produced to communicate shipment distribution, route patterns, and performance comparisons in a business-readable format.

6. Key Performance Indicators

KPI	Observed result
Total shipment records	10,194
Average shipping lead time	≈ 1,320.84 days
Lead-time minimum	≈ 904 days
Primary shipment mode	Standard Class
Largest shipment region	Pacific

Interpretation: The average lead-time figure is exceptionally high for normal parcel shipping and should therefore be treated as a data-quality signal rather than a confirmed operational benchmark.

7. Regional Shipment Analysis

Region	Shipments	Share of records
Pacific	3,253	31.91%
Atlantic	2,986	29.29%
Interior	2,335	22.90%
Gulf	1,620	15.89%

The Pacific region has the highest shipment volume at 3,253 records, accounting for roughly 31.9% of the dataset. Atlantic follows with 2,986 records. Together, Pacific and Atlantic represent approximately 61.2% of all shipment records, making these regions important areas for operational monitoring.

8. Shipping Mode Analysis

Ship Mode	Shipments	Share
Standard Class	6,120	60.04%
Second Class	1,979	19.41%
First Class	1,548	15.19%
Same Day	547	5.37%

Standard Class is the dominant shipment mode, representing approximately 60% of records. Second Class is the next largest category. The smaller Same Day and First Class categories can be analyzed separately when evaluating premium-service demand and route requirements.

9. Factory-to-Customer Route Analysis

Route-level aggregation was used to evaluate shipment volume and average lead time for factory-to-customer combinations. The analysis also included state-level summaries and a top-route visualization to identify routes with comparatively high shipment activity.

Because the underlying shipping dates produce unusually large lead-time values, route efficiency should currently be interpreted primarily through relative comparisons—such as which routes have higher or lower calculated lead times—rather than through absolute service level targets.

10. Key Findings

• The dataset contains 10,194 shipment records suitable for exploratory logistics analysis.

• Pacific is the largest region by shipment volume, followed by Atlantic.

• Standard Class accounts for the majority of shipment records.

• Factory-to-customer route aggregation provides a useful structure for identifying high-volume routes and geographic areas for investigation.

• State and regional comparisons can help prioritize operational review.

• The unusually high calculated lead times are the most important data-quality limitation and should be validated before operational use.

11. Business Recommendations

• Validate the original Order Date and Ship Date fields to confirm that they represent the intended business events.

• Investigate high-volume routes first because improvements on these routes could have the largest operational impact.

• Monitor Pacific and Atlantic routes closely due to their combined share of shipment activity.

• Segment premium shipping modes such as Same Day and First Class for separate service-level analysis.

• Introduce distance or geographic coordinates in future versions to distinguish true route inefficiency from data-date anomalies.

• Use the analysis as a monitoring and investigation framework rather than a direct service-level benchmark until the date-quality issue is resolved.

12. Data Quality & Limitations

The calculated shipping lead times are unusually high. The available records include order dates and ship dates that can be separated by multiple years, producing lead times of roughly 900 days or more in the observed data. This is inconsistent with typical operational shipping timelines.

> Possible explanations include historical data, simulated data, mismatched date fields, or source-data quality issues. The project therefore avoids presenting the calculated lead-time values as verified real-world delivery performance. Validation against source records is recommended before any operational decision is made.

13. Conclusion

The Nassau Candy shipping-route analysis establishes a structured framework for evaluating factory-to-customer logistics performance. It combines data validation, feature engineering, route aggregation, KPI measurement, regional analysis, and shipping-mode analysis to provide a clear view of shipment patterns.

The analysis identifies Pacific as the highest-volume region and Standard Class as the dominant shipping mode, while also demonstrating how route-level analysis can support operational investigation. The most important next step is validation of the source dates underlying the shipping lead-time calculation.

14. Future Scope

• Develop an interactive Streamlit dashboard.

• Add geographic distance and mapping features.

• Build route-cost and route-time optimization models.

• Create predictive models for shipping-delay risk after date validation.

• Add automated KPI monitoring and anomaly detection.

• Integrate validated logistics data for near-real-time reporting.

15. Tools & Technologies

Tool	Purpose
Python	Core analysis and feature engineering
Pandas	Data cleaning, transformation, and aggregation

Tool	Purpose
NumPy	Numerical operations
Matplotlib	Charts and visualizations
Seaborn	Statistical visualization
Jupyter Notebook	Interactive analysis and documentation
Streamlit	Planned interactive dashboard

16. Project Information

Project: Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor
Author: Nikhil Chikte
Repository: nassau-candy-shipping-analysis
Status: Analytical phase completed; dashboard and deployment planned.